Object-Orientation Concepts

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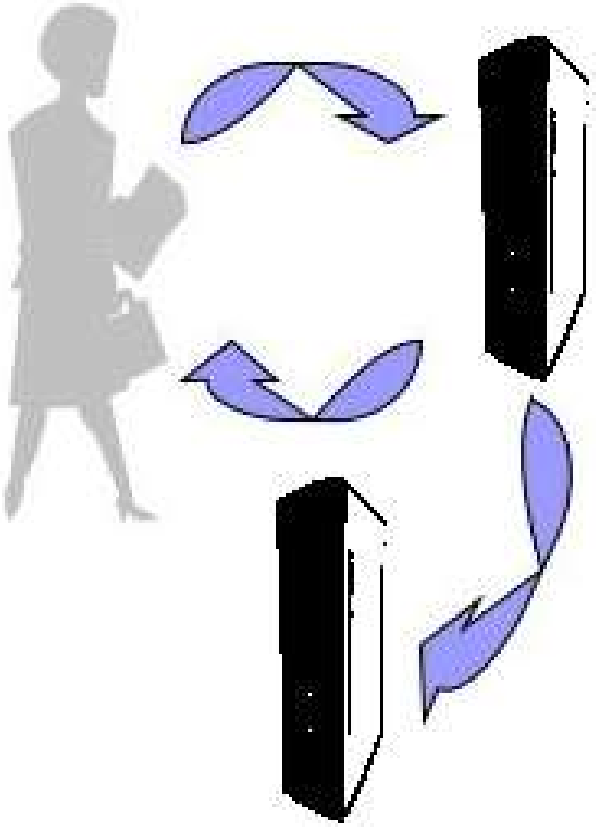
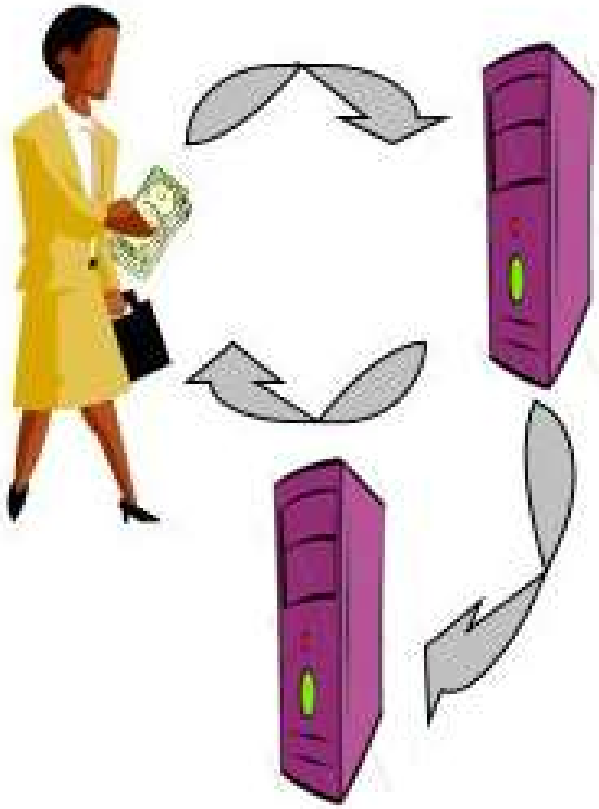
Function Oriented Design vs. Object Oriented Design

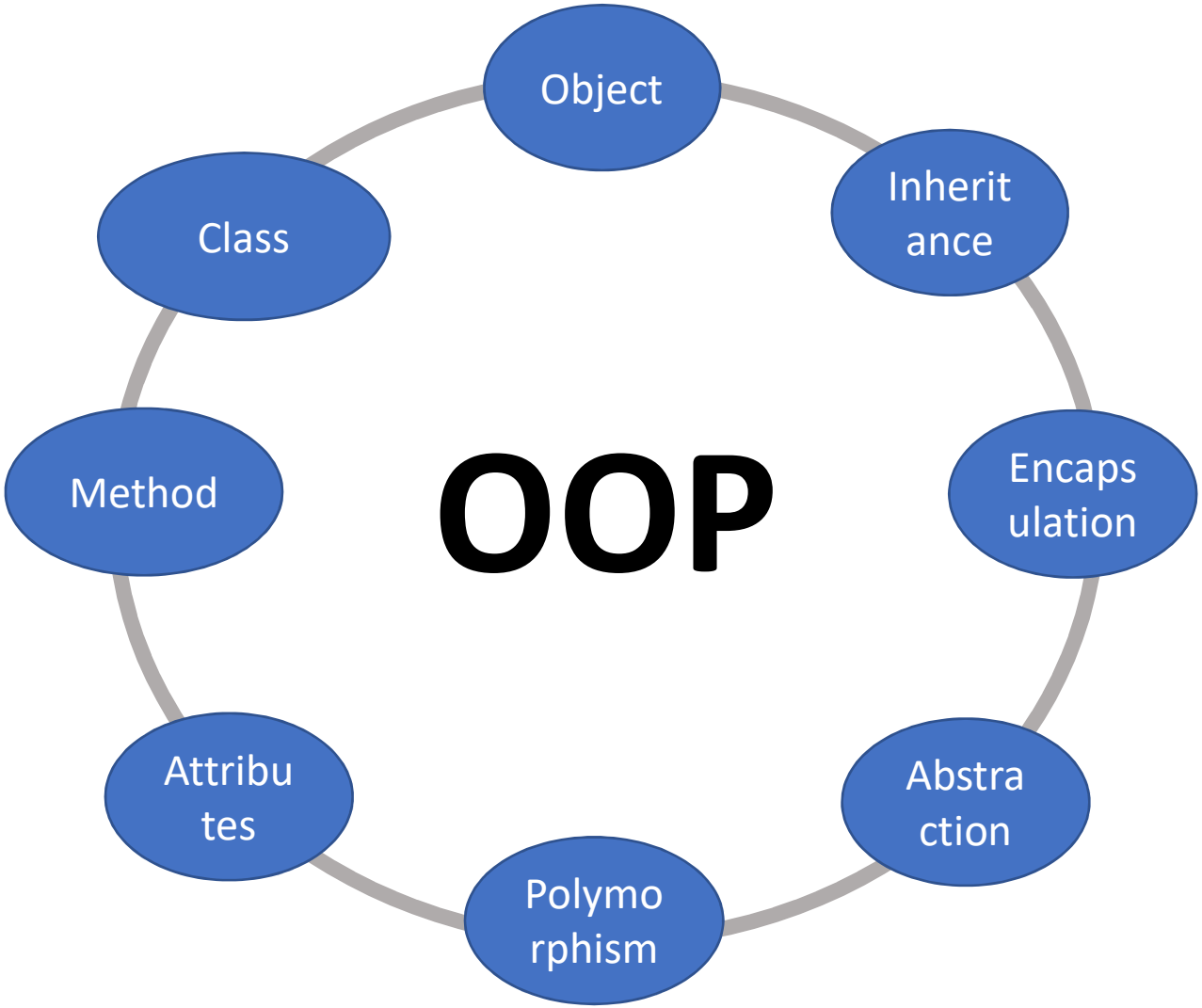


Function Oriented
Registration, **Issue**, **Return**



Object Oriented
Administrator, **Member**, **Book**

Object-Orientation Concepts



UML
Unified Modelling Language

What is a model?

- A **model** is an **abstraction of the real problem** (or situation), and is constructed by leaving out unnecessary details.
  - This **reduces the problem complexity** and **makes easy to understand the problem**.
- A **model** can also be **described as a simplified version of real system**.
- As a **model presents only some of the features/views** of the real system, **we may need** to have **multiple models for describing complete real system**.
  - For example to describe a real building we may need several models describing the building from different prospective such as:
    1. External view of the building
    2. Floor plan
    3. Electrical configuration etc..
- A model can be graphical/textual/mathematical/program code based.

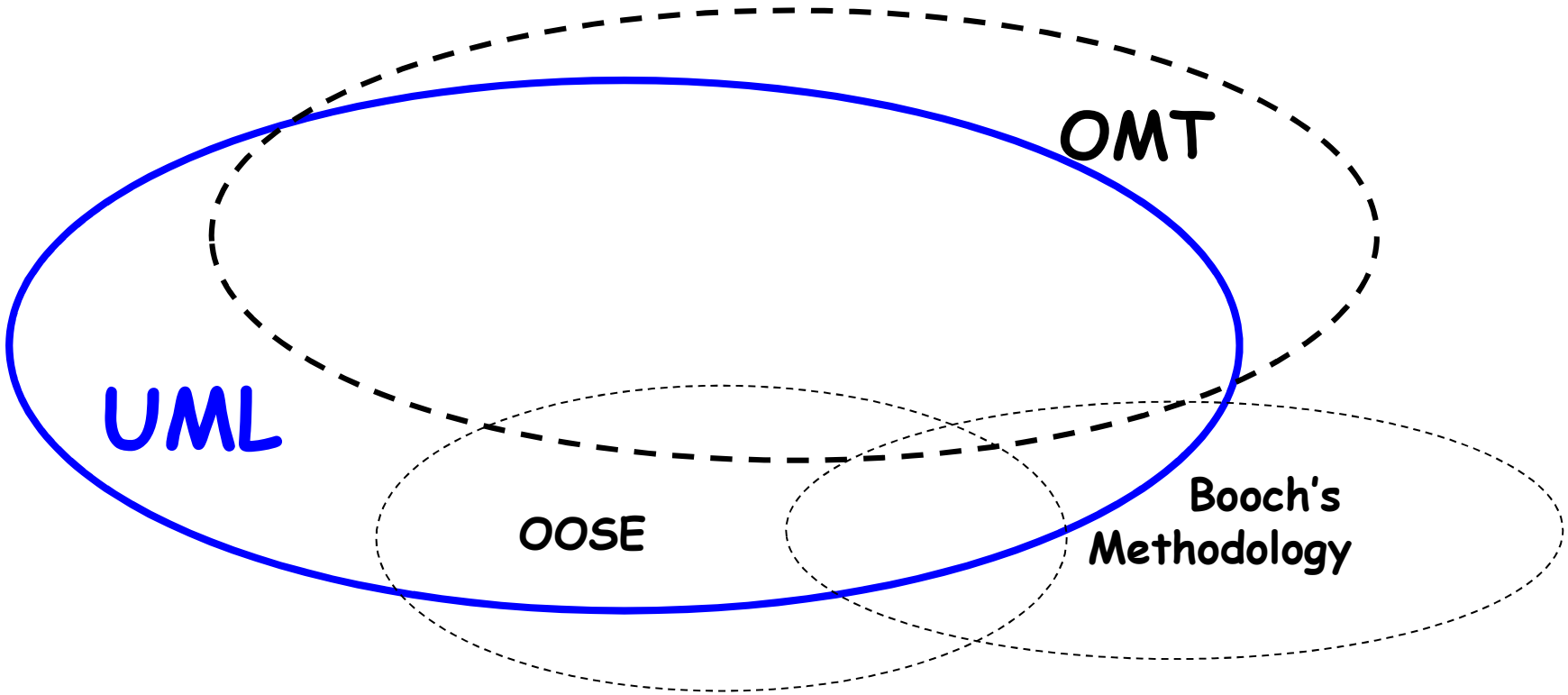
What about model in the context of S/W development?

- **UML is a graphical modeling tool for OO software development**. Primarily **different types of diagrams are used to describe different views of the system**. Textual explanations also accompany the diagrams.
- The models basically useful for the different phases of s/w development such as:
  - Analysis
  - Specification
  - Design
  - Coding
  - Visualization and understanding of the implementation
  - Testing etc..
- **A model is expected to describe only those features of the system depending on the purpose for which it is constructed**.

UML

- **UML is based Principally on the following :**
  - **OMT (Object Modelling Technique)** [Rumbaugh 1991]
  - **Booch's methodology**[Booch 1991]
  - **OOSE** [Jacobson 1992]
  - **Odell's methodology**[Odell 1992]
  - **Shlaer and Mellor** [Shlaer 1992]

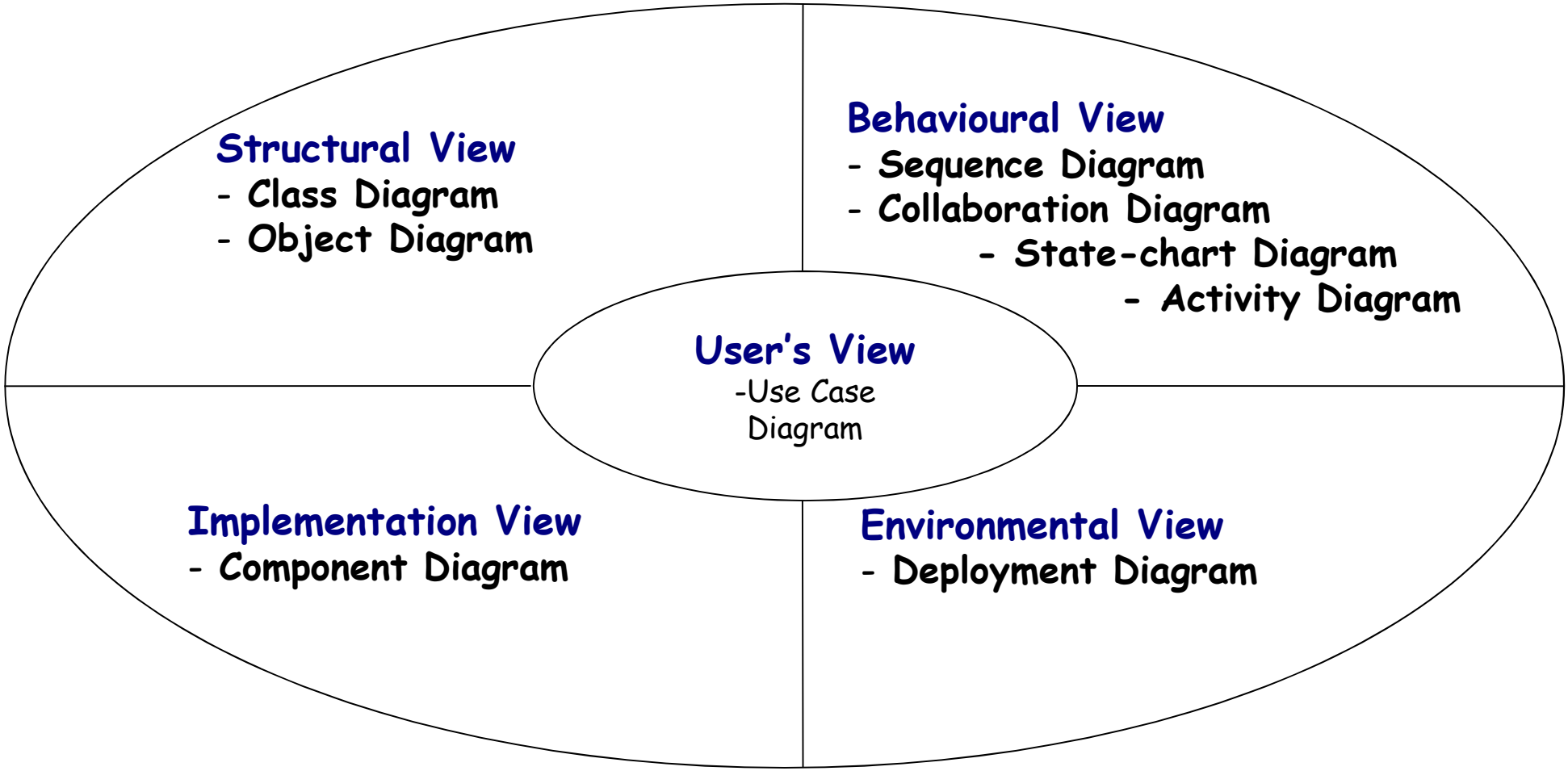
Different Object Modelling Techniques in UML



UML

- **UML** can be used to construct **nine** different types of **diagrams** to capture **five** different **views of a system**.
- **Each diagram focuses on some specific aspect** and ignores the rest.
- UML diagram can capture following views (models) of a system
  - **1 User's view**
    - Use Case Diagram
  - **2 Structural view**
    - Class Diagram
    - Object Diagram
  - **3 Behavioural view**
    - Sequence Diagram
    - Collaboration Diagram
    - State-chart Diagram
    - Activity Diagram
  - **4 Implementation view**
    - Component Diagram
  - **5 Environmental view**
    - Deployment Diagram

UML Diagrams



Diagrams and views in UML

Different views

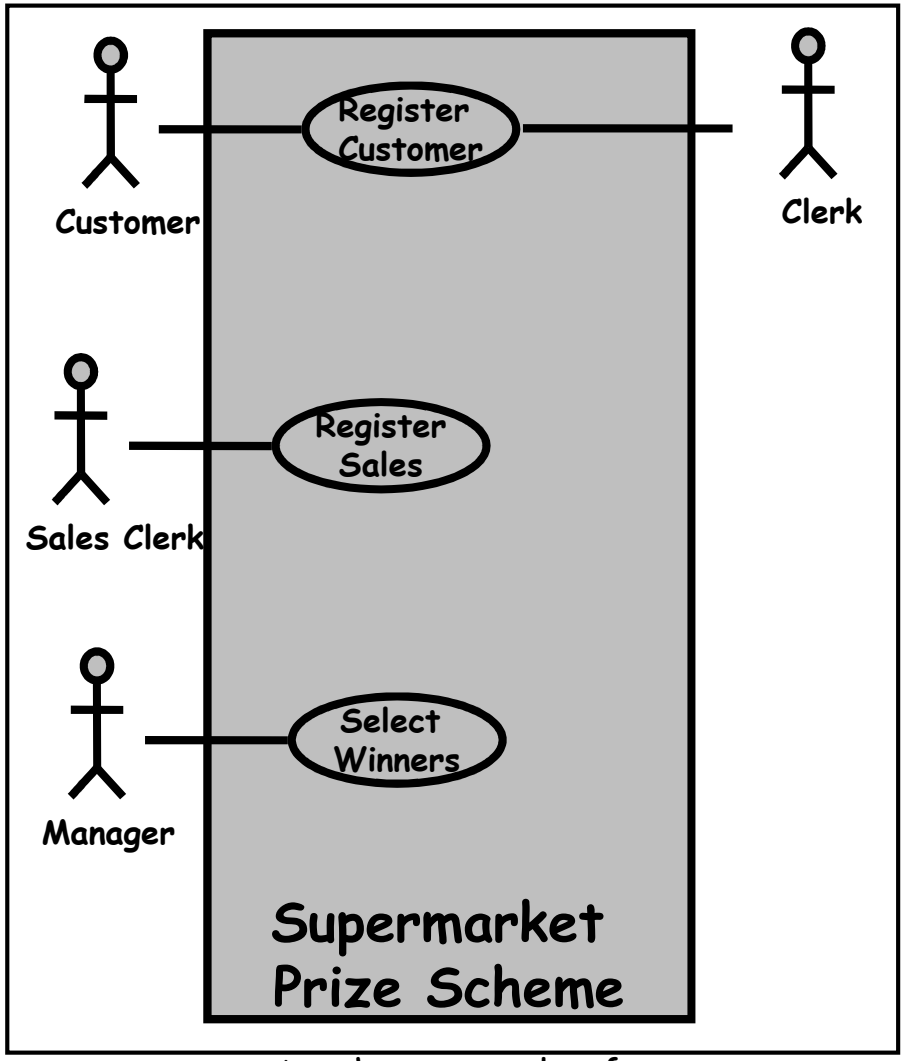
- **Users' view**
  - **This view captures** the view of the system in terms of the **functionalities offered by the system to its users**.
  - It is a **black box view**: internal structures, dynamic behavior of different components, implementation details are not considered.
  - This is considered as the **central view** and **other views are required to confirm this view**.
- **Structural view**
  - It **defines the structure** of the system, **in terms of kind of objects (classes)** important for understanding of the working of the system, and for its implementation.
  - It is called as the **static model**, as the structure of a system do not vary with time.
- **Behavioral view**
  - It captures the **how objects interact with each other in time** to realize system behavior. As it **captures time dependent dynamic behavior** of the system, it constitutes the dynamic model of the system.

Different views

- **Implementation view**
  - This view **captures important components** of the system such as **GUI**, **middle wire** and **data base** and **their interdependencies**.
- **Environmental view**
  - This view captures **how different components are implemented in different pieces of hardware**.
- Usually **all the views are not necessary for every software** project.
- **For a simple system**, **use case model**, **class diagram** and **one of the interaction diagram** from **behavioral view** is sufficient.
- As we **do not use all word in a dictionary** while **writing a poem**, we **also do not need all the UML diagrams** while **modeling a system**.

How to Represent Use Cases?

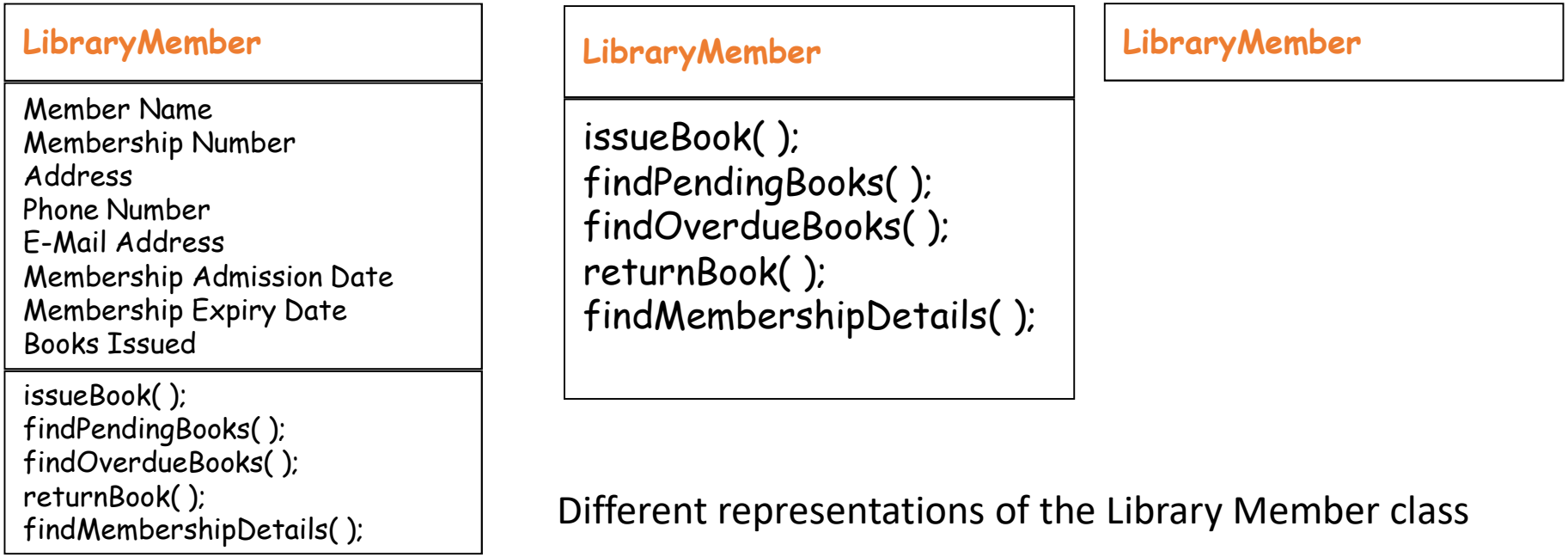
- In addition to main line sequence and alternate scenarios the use case test description may include:
  - **Contact person**
    - With whom use case was discussed
  - **Actors**
    - Identification of the actors will be useful for implementation
  - **Preconditions**
    - If any before execution of the use case
  - **Post-conditions**
    - What will be the state of the system after completion of the use case
  - **Non-functional requirements**
    - Platform of execution, response time etc..
  - **Exceptions error situations**
    - Domain errors, users access right, error in input field etc…
  - **Sample dialog**
    - Example illustrating use case
  - **Specific user interface requirements**
    - Forms to be used, screen shots etc…
  - **Document references**



Another example of Use case model

Class Diagram

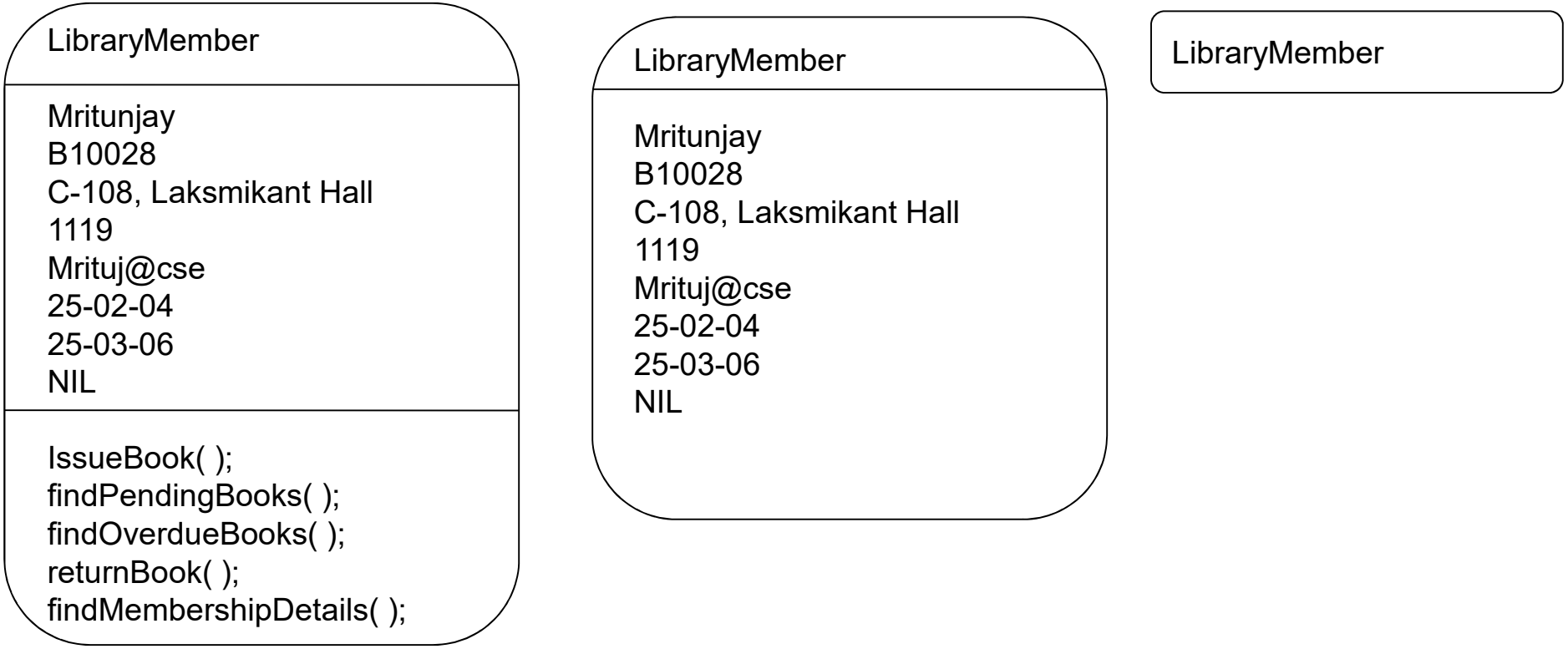
- **Class diagram describes static structure of the system.**
- **It consists of**
  - **a number of class diagrams (represent features: attributes and operations/methods) and**
  - **relationship (generalization, aggregation, association and various kind of dependencies) among them.**



Different representations of the Library Member class

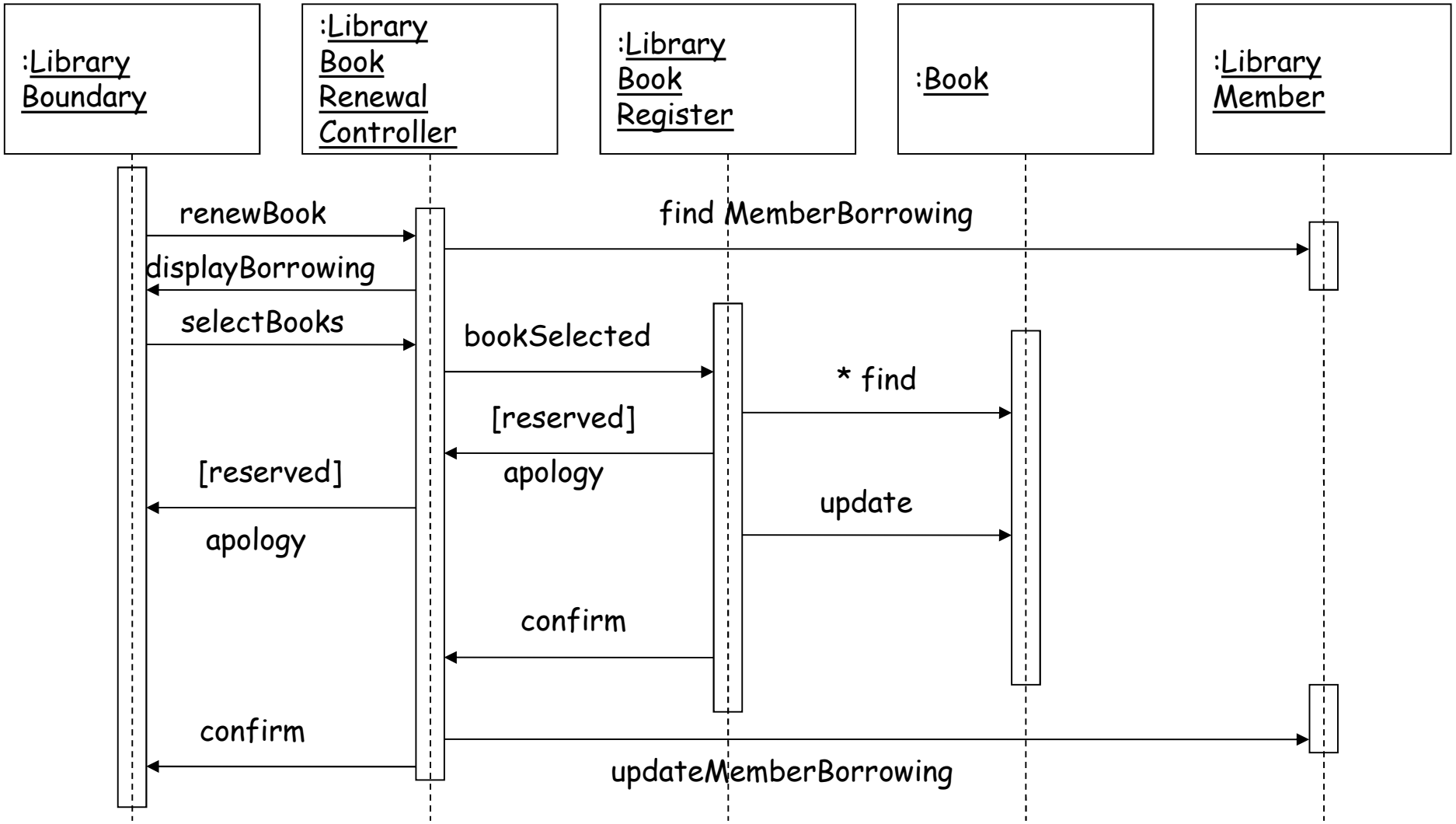
Object diagram

- **Object diagram** shows the **snapshot of the objects** in a system at a point of time**.** It shows **instances of a class and** therefore also **called as instance diagram.**



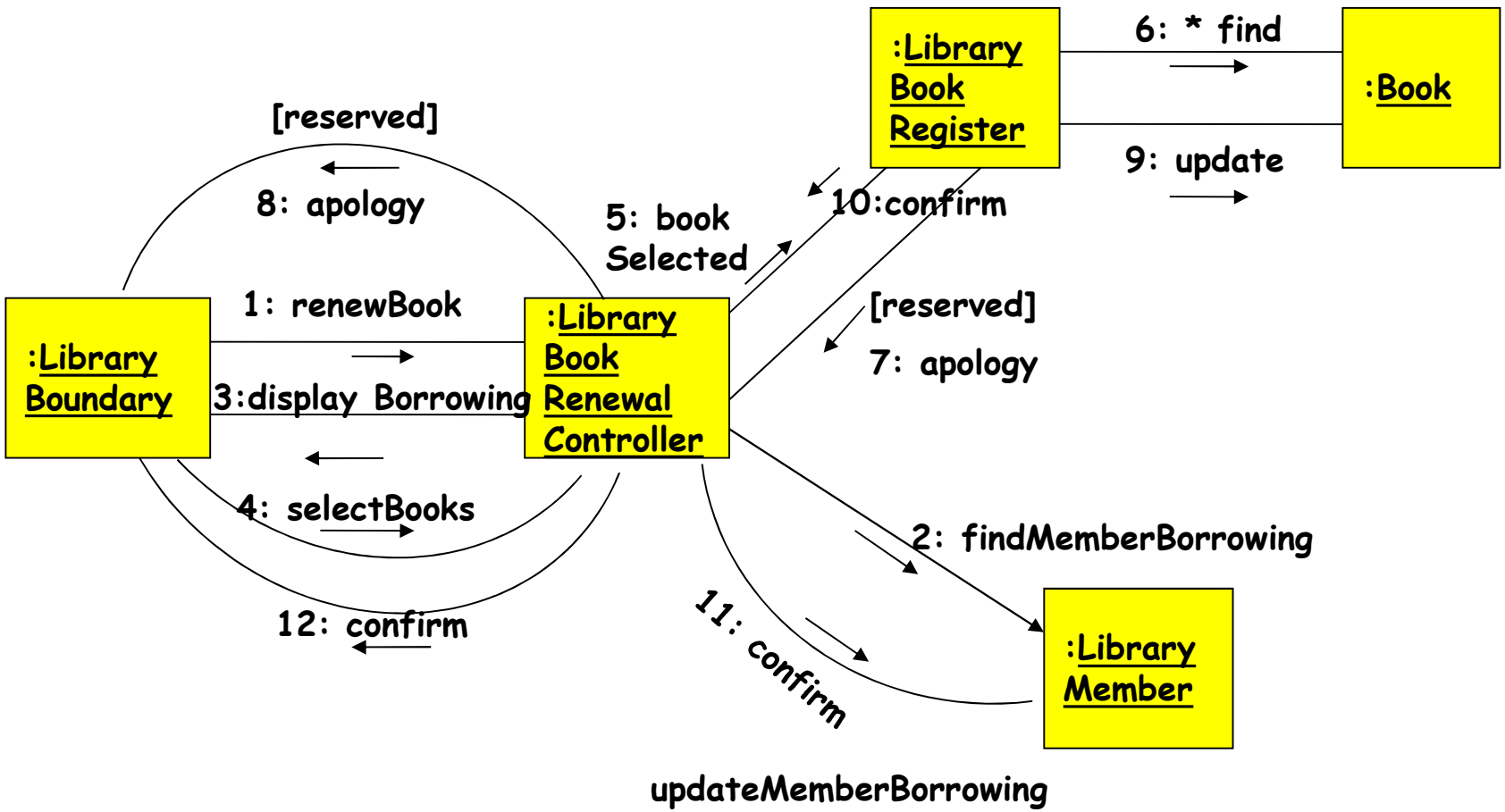
Different representations of the LibraryMember object

Sequence Diagram



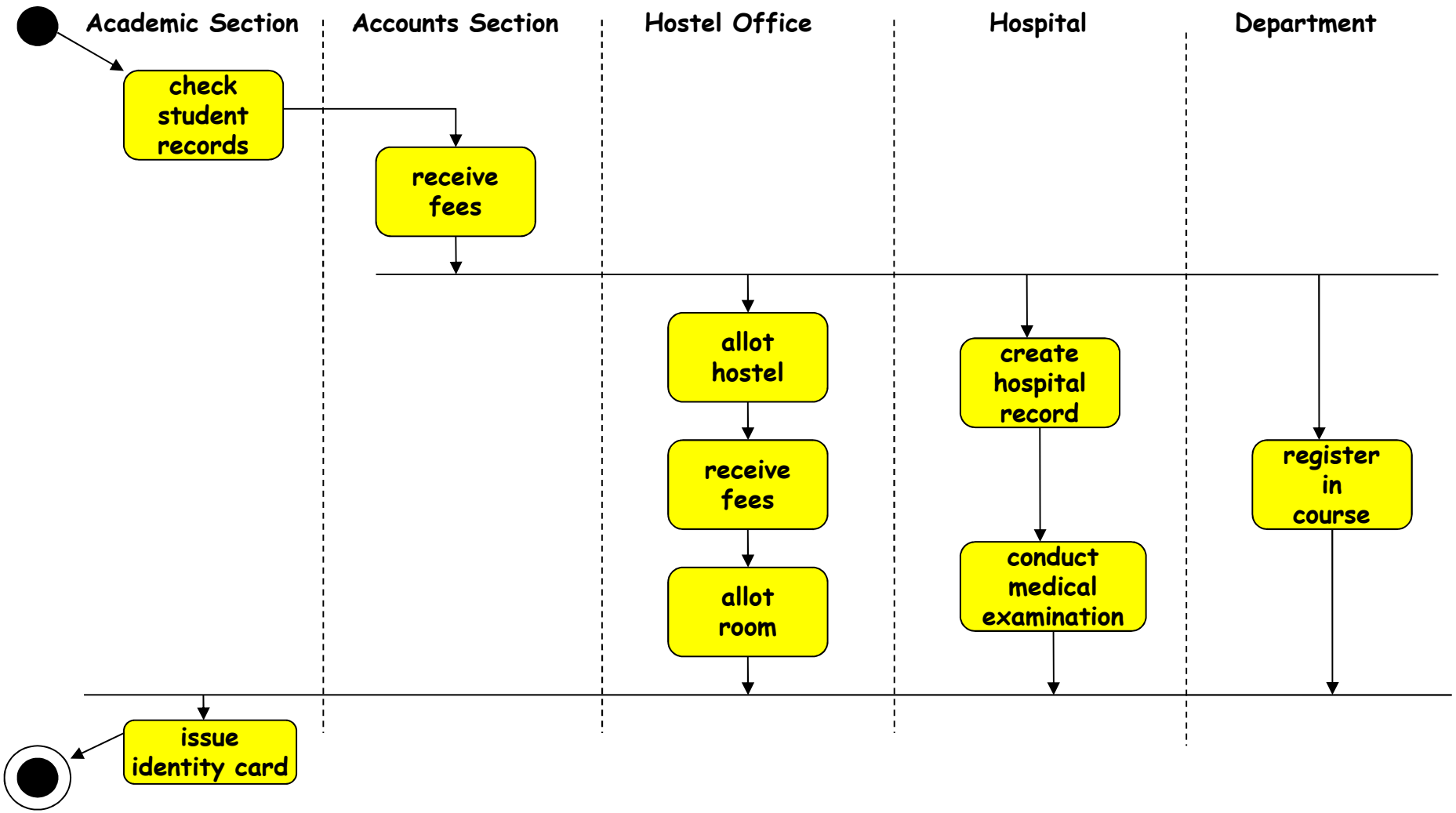
Sequence Diagram for the renew book use case

Collaboration Diagram



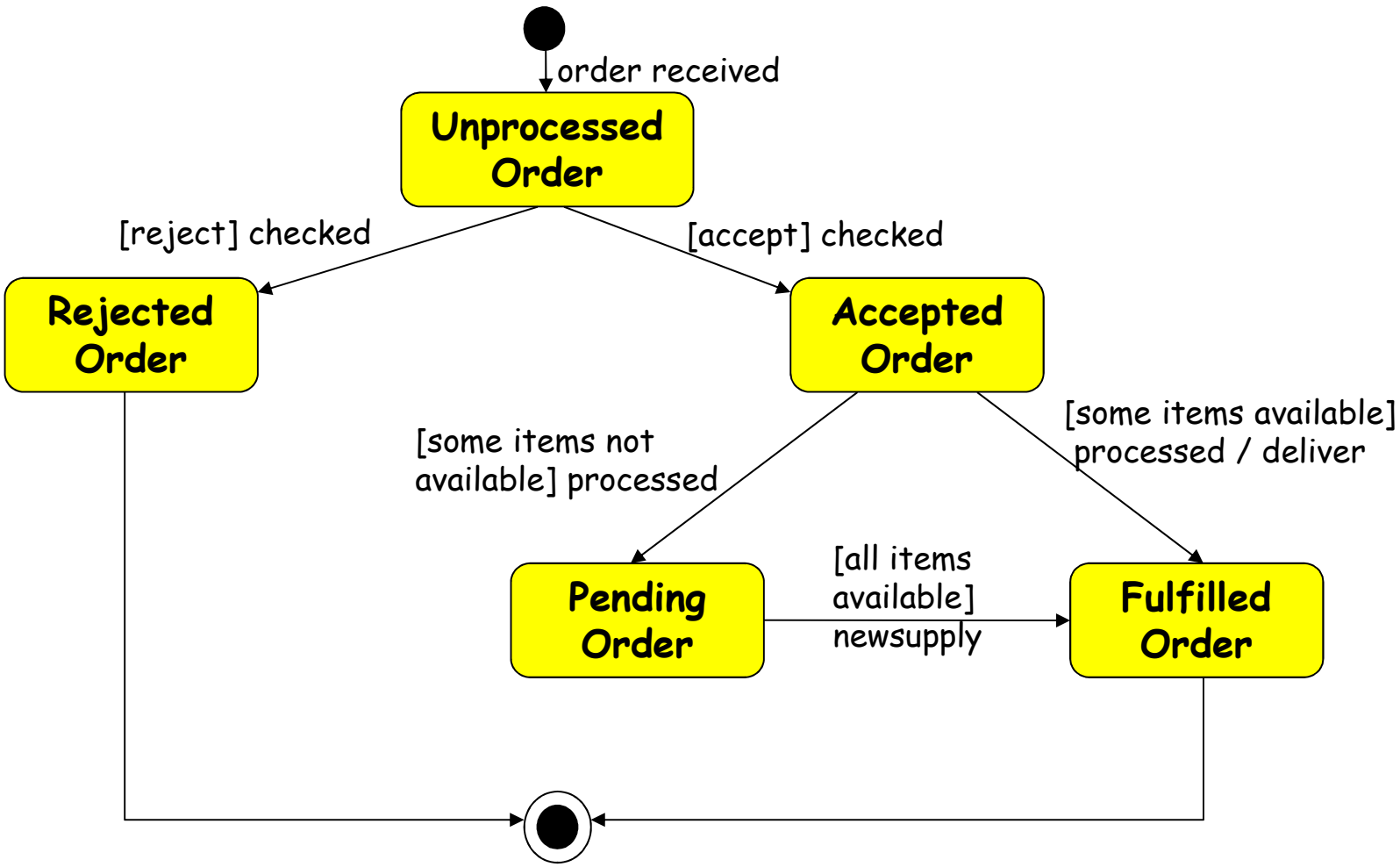
Collaboration Diagram for the renew book use case

Activity Diagram



Activity diagram for student admission procedure

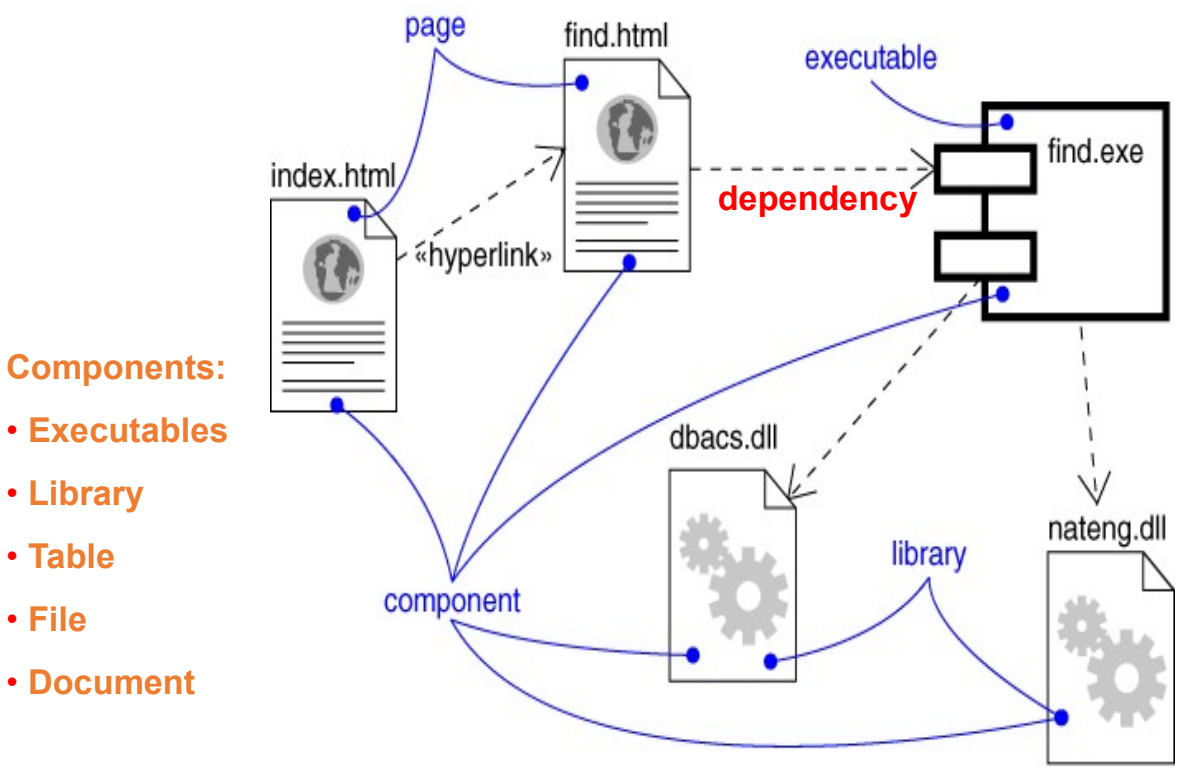
State Chart Diagram

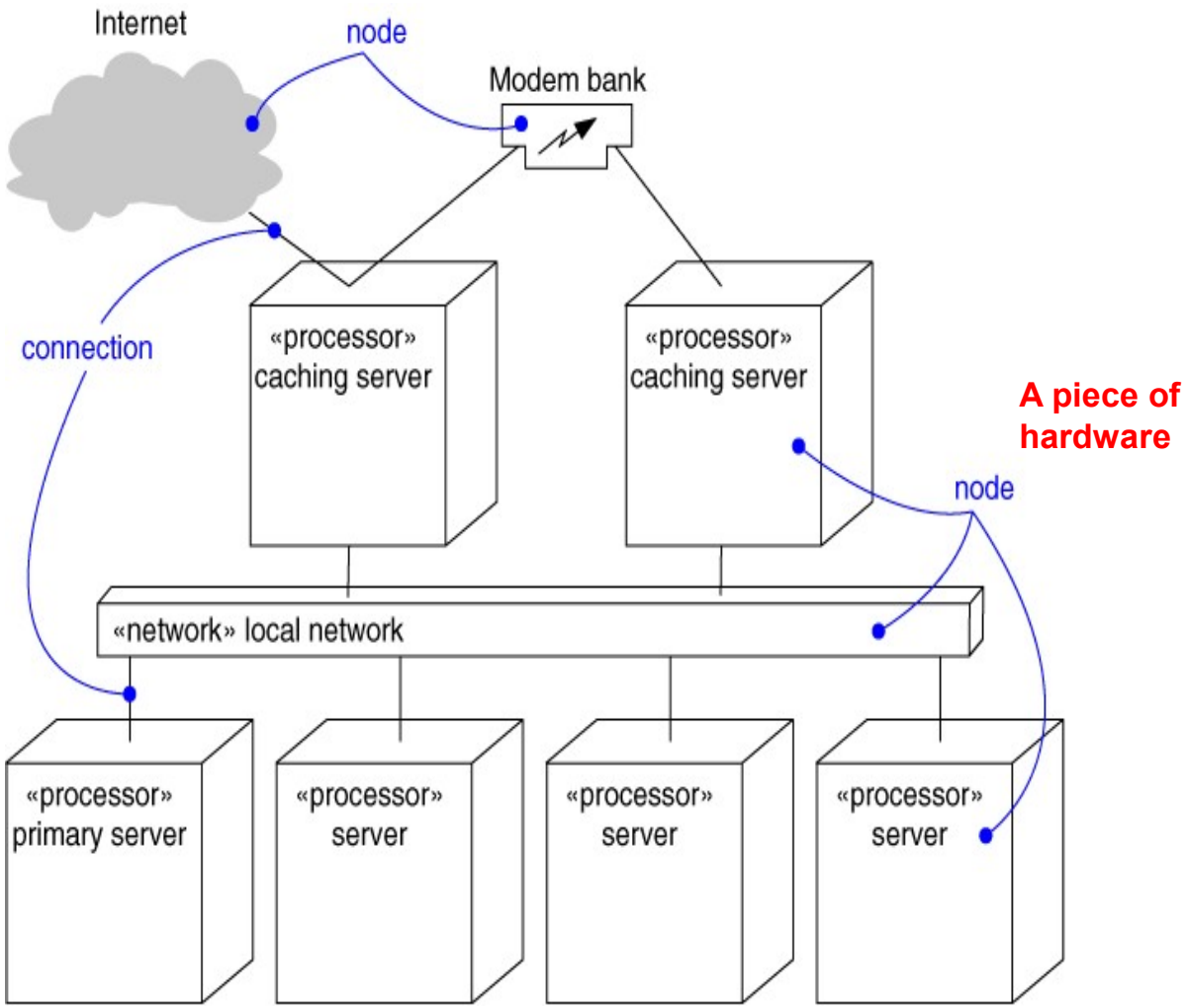


Example: State chart diagram for an order object in an inventory mgmt system.

Component Diagram

- Captures the physical structure of the implementation (code components)



Deployment Diagram

- Captures the **topology of a system's hardware**